Progress Report: How Foveated Vision Shapes Cognitive Maps in Navigation Agents

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CS-503 Visual Intelligence — Progress Report (Updated: April 12, 2026)

I. Overview

Research question: Does non-uniform (foveated) visual input reshape the spatial representations learned in a navigation agent's recurrent memory, compared to blind, uniform-resolution, and matched-compute baselines?

Three hypotheses:

H1 Compensatory memory: Foveated agents develop longer-horizon spatial memory to compensate for degraded peripheral vision.

H2 Representational divergence: Different visual inputs produce qualitatively different representational geometries, not just different accuracy levels.

H3 Epistemic gaze: A learned-gaze agent directs fixation toward memory-weak regions, driven by task pressure alone.

Setup: All agents are trained with DD-PPO [1] on PointGoal navigation in Habitat [2], using Gibson-0+ (411 scenes) + MP3D train (61 scenes). All share an identical 3-layer LSTM (512-d hidden) and the Wijmans non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar). Five conditions vary only the visual input:

- **Blind:** No visual input (replicates Wijmans et al. [3]).
- **Uniform:** Full-resolution 256×256 RGB via ResNet-18.
- **Foveated-fixed:** Eccentricity-dependent Gaussian blur ($\sigma_{\max} = 8$, quadratic falloff), gaze locked at image centre.
- **Matched-compute:** Low-resolution uniform RGB, total pixel budget exceeding foveated effective budget (controls for information volume).
- **Foveated-learned:** Same blur as foveated-fixed, but gaze position predicted by an MLP from the LSTM hidden state (tests H3).

Architecture note: Wijmans et al. [3] trained only blind agents; they did not train a sighted variant. The DD-PPO sighted agent they reference (from [1]) uses ResNet-50 with depth input. We use ResNet-18 with RGB to keep the model smaller and because our research question requires training multiple visual conditions; the matched-compute baseline controls for total visual information regardless of encoder size.

Analysis: Freeze weights → collect per-step LSTM hidden states → linear probing (Ridge regression, episode-level splits) for spatial variables → cross-condition CKA and probe transfer → behavioral tests (shortcut discovery, memory transplant) → PCA/t-SNE visualization.

II. Training Status

Training targets are set to ensure convergence rather than a fixed frame budget.

Table I
TRAINING STATUS PER CONDITION (AS OF APRIL 12, 2026).

Condition	Frames	Target	Success	SPL	Status
Blind	341M	250M	90.7%	0.550	**Complete**
Uniform	206M	400M	97.6%	0.805	Training
Foveated	139M	250M	98.3%	0.823	Training
Matched	203M	250M	92.9%	0.582	**Complete**
Fov-learned	15M	250M	—	—	Training

Design note: Uniform is trained to 400M (vs. 250M for others) because its ResNet-18 encoder requires more samples to converge. We match convergence level rather than frame count to ensure fair cross-condition comparison.

III. Probing Results (Phase 1 – Baseline)

Full 13-experiment probing analysis has been completed for all four conditions using their best available checkpoints (blind: ckpt.19, uniform: ckpt.33, foveated: ckpt.21, matched: ckpt.30). All probes use full 512-d hidden states (no PCA).

A. Global GPS and Compass Probes (1b)

Table II
GLOBAL LINEAR PROBE ACCURACY (RIDGE, $\alpha = 10$, EPISODE-LEVEL TRAIN/TEST SPLIT). HIGHER R^2 = BETTER SPATIAL ENCODING.

	GPS R^2	GPS MAE	Compass R^2	Compass MAE
Blind	**0.899**	0.020m	**0.878**	0.8°
Foveated	0.841	0.034m	0.715	2.1°
Matched	0.832	0.033m	0.607	1.3°
Uniform	0.614	0.035m	0.423	1.6°

Key observations:

- Blind dominates baseline probes—GPS/compass are directly in the input, so the LSTM trivially encodes them. Replicates Wijmans et al. [3].
- Among sighted conditions, **foveated ranks highest** on both GPS (0.841) and compass (0.715).
- Uniform is weakest (GPS 0.614)—the full visual stream may act as a crutch, reducing reliance on memory.

B. Distance-to-Goal and Control Tasks (1c, 1e–f)

All selectivity values are well above zero, confirming probes capture genuine spatial structure, not memorization artifacts.

C. Multi-Layer Comparison (1d)

For the blind agent, layer 0 hidden state (h) achieves the highest GPS R^2 (0.958), while layer 1 cell state (c) is a close second (0.962). The top LSTM layer (layer 2, h) matches the default probe target (0.899). Spatial information is distributed across layers but concentrated in layers 0–1.

Table III
DISTANCE-TO-GOAL PROBE AND SELECTIVITY (HEWITT & LIANG CONTROL).

	Dist-to-goal R^2	GPS selectivity	Compass selectivity
Blind	**0.984**	**0.911**	**0.890**
Foveated	0.953	0.880	0.751
Matched	0.977	0.871	0.701
Uniform	0.959	0.763	0.543

IV. PROBING RESULTS (PHASE 2 – H1: COMPENSATORY MEMORY)

A. Path-History Probe (2c)

This probe decodes the agent's position at k steps *in the past* from the current hidden state, testing how far back spatial memory extends.

Table IV
PATH-HISTORY PROBE: R^2 FOR DECODING GPS AT LAG k FROM CURRENT HIDDEN STATE. HIGHER = LONGER WORKING MEMORY.

	lag-0	lag-1	lag-2	lag-3	lag-4	lag-5
Blind	0.899	0.905	0.867	0.748	0.536	0.536
Foveated	0.841	0.862	0.826	0.737	**0.685**	**0.629**
Matched	0.832	0.841	0.779	0.581	0.475	0.445
Uniform	0.614	0.555	0.393	0.386	0.008	−0.105

This is the strongest finding so far. The foveated agent retains spatial information over longer temporal horizons than any other condition:

- At lag-5, foveated R^2=0.629 vs. blind 0.536, matched 0.445, uniform −0.105.
- Foveated surpasses even blind at lags 4–5, despite blind having GPS directly in its input.
- Uniform memory decays rapidly and becomes uninformative by lag-5.

This is **direct evidence for H1 (compensatory memory)**: degraded peripheral vision forces the foveated agent to build a richer temporal trace of where it has been.

The foveated–matched comparison (0.629 vs. 0.445 at lag-5) isolates the effect of *spatial non-uniformity* from total information volume: it is the uneven quality of the foveated sensor, not simply having less visual information, that drives compensatory memory.

B. Visited-Region Probe (2d)

Table V
VISITED-REGION PROBE: CAN THE HIDDEN STATE PREDICT WHICH SPATIAL CELLS THE AGENT HAS VISITED?

	R^2	Binary accuracy
Blind	**0.834**	99.8%
Uniform	0.826	99.8%
Matched	0.808	99.8%
Foveated	0.801	99.8%

All conditions track visited regions near-perfectly (>99.8% binary accuracy). R^2 values are similar, suggesting this is a basic capability shared by all agents regardless of visual input.

C. Occupancy Map Reconstruction (2e)

Table VI
OCCUPANCY DECODING FROM HIDDEN STATES (20×20 GRID).

	R^2	Binary accuracy
Blind	−0.144	59.6%
Matched	−0.379	58.1%
Foveated	−0.397	55.2%
Uniform	−0.300	56.3%

Occupancy decoding is weak across all conditions (all $R^2 < 0$). None of the agents build detailed layout maps in their hidden states. This may require more episodes, a finer grid, or a non-linear decoder to detect.

D. Place Cells (5a)

Table VII
PLACE-CELL-LIKE UNITS (SPATIAL INFORMATION > 1 BIT) OUT OF 512 HIDDEN UNITS.

	Place cells (>1 bit)
Foveated	**475**/512
Blind	473/512
Uniform	473/512
Matched	461/512

Nearly all units carry spatial information across all conditions. Matched has slightly fewer place-cell-like units (461 vs. ∼473).

V. PROBING RESULTS (PHASE 3 – H2: REPRESENTATIONAL DIVERGENCE)

A. Cross-Condition CKA (3a)

Linear CKA measures representational similarity between conditions (1.0 = identical geometry, 0.0 = unrelated). We compute CKA on the top LSTM hidden states (h_layer_2) using matched episodes across all condition pairs.

Table VIII
LINEAR CKA SIMILARITY (TOP-LAYER HIDDEN STATE, $n \approx 1{,}910$ STEPS).

	Blind	Uniform	Foveated	Matched
Blind	—	0.0023	0.0023	0.0011
Uniform		—	**0.0007**	0.0017
Foveated			—	0.0011

Key observations:

- All CKA values are near zero (<0.003), indicating that each condition develops a **fundamentally different representational geometry** for encoding space.
- Blind is the most dissimilar to all sighted conditions—consistent with its reliance on proprioceptive rather than visual features.
- The closest pair is uniform–foveated (0.0007), yet still near zero, showing that even sighted agents with different visual sampling diverge strongly.

B. Probe Transfer (3b)

We train a GPS probe on one condition's hidden states and evaluate it on another's. Self-accuracy (∼0.96–0.97) confirms probes work; cross-condition transfer tests whether spatial information is encoded in a shared format.

Table IX
PROBE-TRANSFER R^2: GPS PROBE TRAINED ON ROW, TESTED ON COLUMN.

Train \ Test	Blind	Uniform	Foveated	Matched
Blind	**0.973**	−9.72	−2.71	−15.72
Uniform	−7.26	**0.973**	−13.53	−11.47
Foveated	−11.25	−5.12	**0.963**	−10.66
Matched	−5.70	−11.74	−15.52	**0.959**

Cross-condition transfer is **catastrophically negative** in all cases, confirming that spatial information is not encoded in a shared linear subspace across conditions. Combined with near-zero CKA, this provides **strong evidence for H2**: visual input type fundamentally shapes the representational geometry used to encode space, not just the amount of spatial information stored.

VI. Behavioral Test: Shortcut / Cognitive Map

The shortcut discovery test [4] evaluates whether an agent's accumulated spatial experience *functionally* aids navigation—the behavioral signature of a cognitive map. For each scene, we run 10 episodes sequentially in two conditions:

- **Reset** (standard): LSTM hidden state zeroed between episodes.
- **Persistent** (cognitive-map): hidden state carries over, allowing spatial knowledge to accumulate across episodes within the same scene.

If persistent > reset, the agent's recurrent representations encode reusable spatial knowledge. Results for the blind agent (ckpt.32, 341M frames) across 20 Gibson scenes (200 episodes total per condition):

Table X
SHORTCUT / COGNITIVE-MAP BEHAVIORAL TEST (BLIND AGENT, 20 SCENES × 10 EPISODES).

Condition	Success rate	SPL
Reset (standard)	**0.78**	**0.432**
Persistent (cogmap)	0.24	0.145
Difference	−0.53	−0.287

Key finding: Persistent memory **catastrophically hurts** the blind agent (−53% success, −0.287 SPL). The within-persistent-condition trend also shows degradation over time (late–early SPL gap: −0.146), confirming that accumulated hidden state progressively corrupts performance.

Interpretation: The blind agent's LSTM representations are *episode-scoped path-integration states*, not reusable cognitive maps. The hidden state encodes "where I am relative to where I started this episode," and carrying it across episode boundaries places the LSTM in an out-of-distribution regime. This refines H1: the compensatory memory we observe via probing (lag-5 R^2=0.629 for foveated) operates *within* episodes, not across them.

Status: Shortcut eval for the three sighted conditions (uniform, foveated, matched) is currently queued on the cluster. These results will reveal whether visual input enables more transferable spatial representations.

VII. Summary of Hypotheses

Table XI
HYPOTHESIS STATUS BASED ON CURRENT RESULTS.

Hypothesis	Status	Key evidence
H1: Compensatory memory	**Supported**	Foveated lag-5 R^2=0.629 (best across all conditions)
H2: Representational divergence	**Supported**	All CKA <0.003; cross-transfe catastrophically negative
H3: Epistemic gaze	In progress	Learned-gaze training at 15M/2
Additional finding:		
Cognitive map (behavioral)	**Negative (blind)** Pending (sighted)	Persistent LSTM −53% succes Eval queued for 3 conditions

VIII. Next Steps

1) **Complete training** (in progress): Foveated (139M→250M, ∼12h), uniform (206M→400M, ∼15h). Blind and matched already complete.
2) **Shortcut eval for sighted conditions** (queued): Uniform, foveated, matched eval jobs submitted. Will reveal whether visual input enables more transferable cross-episode spatial representations than blind.
3) **Foveated-learned training** (in progress, 15M/250M): Config, policy, and slow-gaze approximation implemented. Running on cs-503 QOS (∼5 days to target).
4) **Re-probe after convergence:** Re-run full 13-experiment probing suite on final checkpoints for definitive comparison. Current probing results use mid-training checkpoints.
5) **Visualization:** PCA/t-SNE of hidden states colored by position and condition. Script ready (`scripts/probing/visualize.py`).
6) **Memory transplant test:** Cross-condition hidden state transplant (`scripts/eval/transplant.py`) to test whether spatial representations are interchangeable across conditions (complements CKA/transfer probing with a behavioral test).

IX. Infrastructure and Tooling

- **Cluster:** SCITAS Izar (EPFL), V100 GPUs. cs-503 QOS: max 2 submitted, max 1 running, 7d12h wall-time.
- **Probing suite:** 13 experiments per condition (GPS, compass, distance-to-goal, multi-layer, selectivity, path-history lags 0–5, visited-region, occupancy map, place-cell rate maps). All implemented in `scripts/probing/`.
- **Cross-condition analysis:** Linear CKA + probe transfer (`scripts/probing/analyze_cross.py`).

- **Behavioral tests:** Shortcut discovery with persistent vs. reset LSTM (`scripts/eval/shortcut.py`).
- **Visualization:** PCA/t-SNE + path-history decay plots (`scripts/probing/visualize.py`).
- **Key lesson:** PCA-32 dimensionality reduction severely underestimates probe accuracy (blind GPS R^2 drops from 0.899 to 0.580). All final probes use full 512-d hidden states.

X. Progress Log

Weeks 1–2 (Mar 24–Apr 6): Codebase setup; implemented all 5 visual conditions; launched training for blind, uniform, foveated-fixed, matched-compute on Izar.

Week 3 (Apr 7–12): Probing infrastructure built (13-experiment suite). Discovered PCA-32 artifact, switched to full 512-d. Completed Phase 1 probing for all 4 training conditions. Ran cross-condition CKA (H2 supported). Fixed eval pipeline bugs (stale code path, checkpoint key format). Completed blind shortcut eval—persistent LSTM memory catastrophically hurts (-53% success), showing episode-scoped path integration rather than reusable cognitive maps. Launched foveated-learned training (H3). Submitted shortcut eval for 3 sighted conditions.

Week 4 (Apr 13–): *Planned:* Complete remaining training (foveated, uniform, fov-learned). Collect sighted shortcut results. Re-probe with final checkpoints. Generate visualizations. Run memory transplant. Begin final paper.

References

[1] E. Wijmans, A. Kadian, A. Morcos, S. Lee, I. Essa, D. Parikh, D. Batra, and M. Savva, "DD-PPO: Learning near-perfect pointgoal navigators from 2.5 billion frames," in *International Conference on Learning Representations (ICLR)*, 2020.

[2] M. Savva, A. Kadian, O. Maksymets, Y. Zhao, E. Wijmans, B. Jain, J. Straub, J. Liu, V. Koltun, J. Malik, D. Batra, and D. Parikh, "Habitat: A platform for embodied AI research," in *International Conference on Computer Vision (ICCV)*, 2019.

[3] E. Wijmans, M. Savva, I. Essa, S. Lee, A. Morcos, and D. Batra, "Emergence of maps in the memories of blind navigation agents," in *International Conference on Learning Representations (ICLR)*, 2023.

[4] S. K. Ramakrishnan, E. Wijmans, P. Krähenbühl, and V. Koltun, "Does spatial cognition emerge in frontier models?" in *International Conference on Learning Representations (ICLR)*, 2025.